

OpenAQ Python Training

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Today's agenda (session I)

- Ice-breaking activity (10 mins)
- Overview of the OpenAQ Python SDK (10 mins)
- Setting up the notebook on your computer (10 mins)
- OpenAQ Python SDK tutorials part I: Intro to key resources (20 mins)
- Q&As (10 mins)

Ice-breaking

(using your phone is recommended for this activity!)

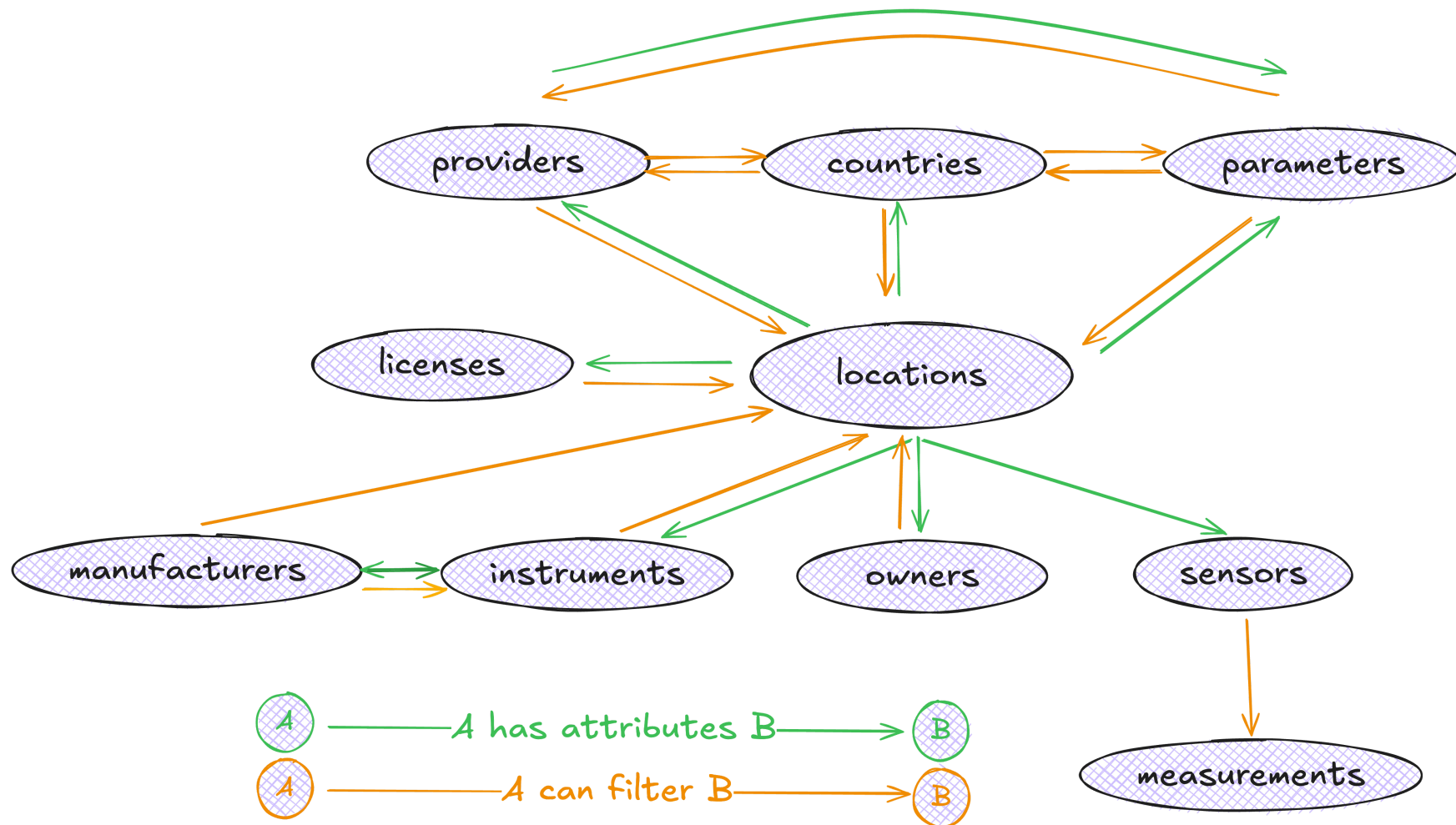
About the Python SDK

- Part of our core tech products
- Allows you to connect to our API and programmatically access air quality data hosted on OpenAQ platform
- Latest version: **1.0.3**

Why use the Python SDK?

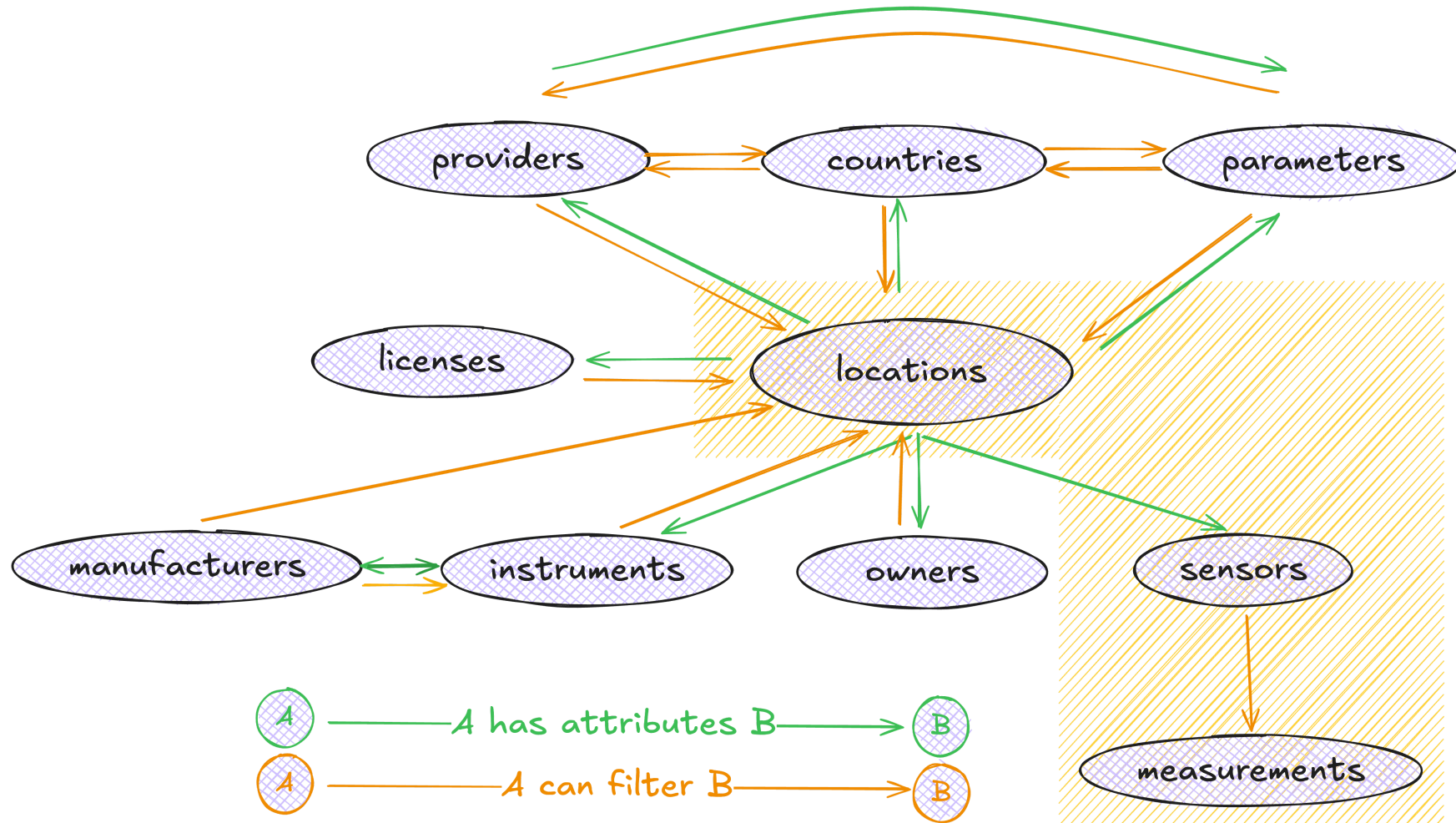
- Accessing air quality data in bulk, from any location and any time
- User-friendly helpers: expressive error handling, preemptive error catching, automatic rate limiting
- More intuitive data access: Python objects-oriented instead of API json responses
- Fully typed: autocomplete, inline documentation

Data models



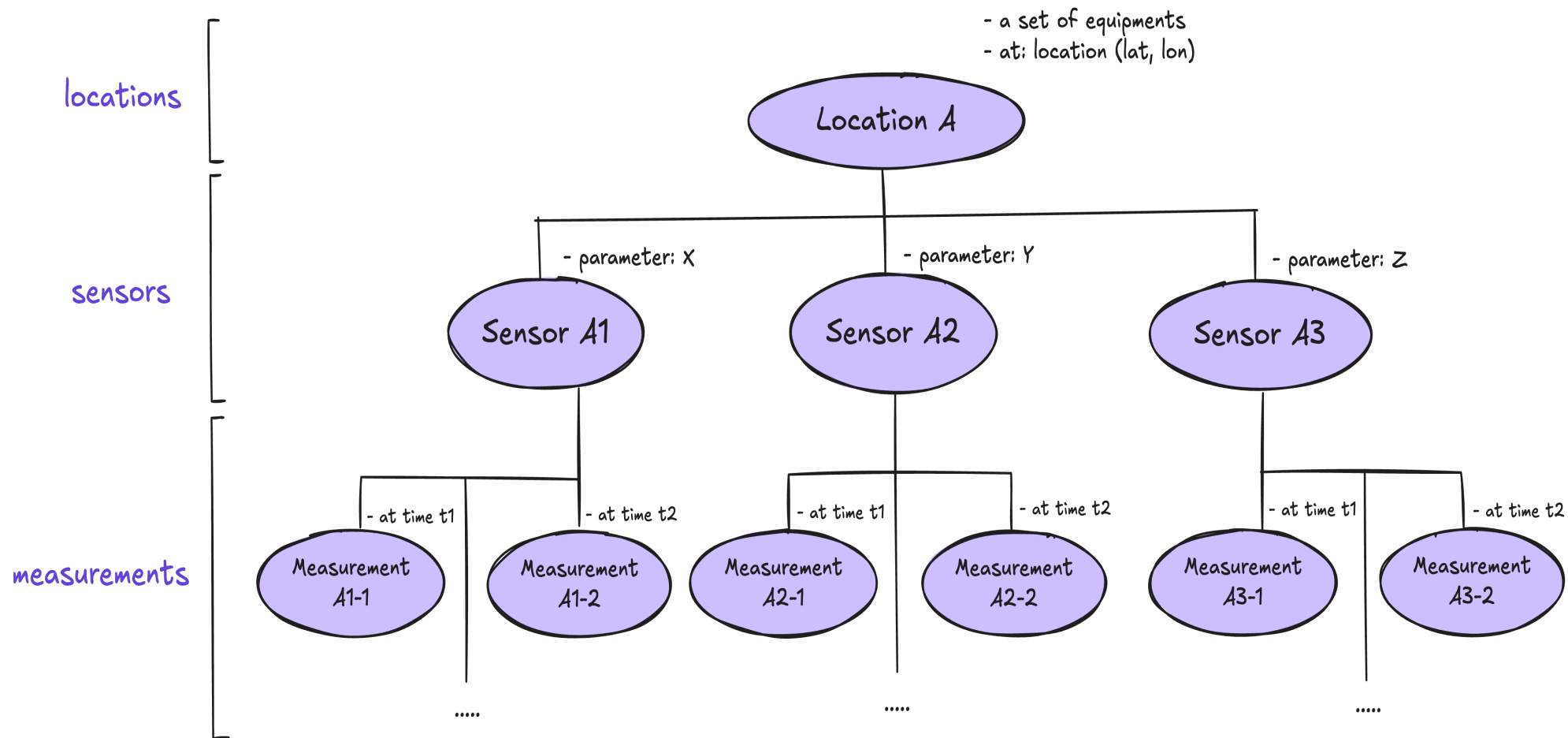
For a full definition of each resource, visit our [Python documentation website](#)

Data models



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Data models



Every Location and Sensor instance gets a unique ID.

Key concepts

API key

The OpenAQ API uses API keys to authenticate requests and manage access.

- control usage and enforce rate limits
- unique and can be requested anew to each users.

Important

Do not share your API key with anyone.

Rate limiting

Our API limits users' usage at:

- 60 requests/minute
- 2,000 requests/hour

Note

From v1.0.0rc1, our Python package supports automatic rate limiting.

Pagination

- By default, a request will return a maximum of 100 results a page. This is adjustable and can be set up to 1,000 results per page using `limit=1000`.
- If a request returns more than 1,000 results, you can loop through the pages until no more data is found.



Tip

Querying the entire dataset and paging can be slow. Adding constraints will help your requests be more performative, especially for volume-intensive resources like Measurements.

```
{'meta': {'name': 'openaq-api',  
  'website': '/',  
  'page': 1,  
  'limit': 1000,  
  'found': '>1000'}}
```

Tutorials

Overview

The tutorial is exercises-driven and works as a primer to OpenAQ Python SDK. Having the code run on your machine as we go is encouraged but optional.

If you want to run the code on your computer as you follow along, by now you should have:

- A [molab account](#)
- Your own [OpenAQ API key](#)

Let's start

Step 1: Go to [this link](#) to create a notebook session for this tutorials

Step 2: Once you're in the link:

- Click “Fork” at the top of the page. You will be prompted to log in to your molab account if you're currently logged out.
- Once you have logged in, you will be automatically redirected to your own forked version of the notebook on the web.
- The notebook is yours to play with and will also be accessible to you via your molab homepage.

Useful links

- [OpenAQ Python SDK documentation](#)
- [OpenAQ API documentation](#)
- [OpenAQ Explorer](#)
- Join [OpenAQ slack channel](#) for questions and discussions anything air quality